

Special topics in logic and security II

Exam solutions — May 16, 2024

Problem 1 — Weight polynomial of $C \subset \mathbb{F}_{11}^3$

Let $C = \{(x, y, z) \in \mathbb{F}_{11}^3 \mid ax + by + cz = 0\}$ with $a, b, c \in \mathbb{F}_{11} \setminus \{0\}$ fixed. The map $\phi : \mathbb{F}_{11}^3 \rightarrow \mathbb{F}_{11}$, $\phi(x, y, z) = ax + by + cz$, is a nonzero linear functional, so $C = \ker \phi$ is a linear subspace of dimension $n - 1 = 2$. Thus $|C| = 11^2 = 121$.

Let A_i be the number of codewords of weight i .

- $A_0 = 1$ (only the zero word).
- $A_1 = 0$: a weight-1 word, say $(x, 0, 0)$ with $x \neq 0$, would force $ax = 0$, impossible since $a \neq 0$. The same argument applies to the other positions.
- A_2 : exactly one coordinate is zero. Take e.g. $z = 0$, $x, y \neq 0$. Then $ax + by = 0$ gives $y = -ab^{-1}x$. For each $x \in \mathbb{F}_{11}^\times$ this yields a unique $y \in \mathbb{F}_{11}^\times$, so 10 codewords. By symmetry the same count holds for the cases $y = 0$ and $x = 0$, totalling $A_2 = 30$.
- $A_3 = 121 - 1 - 0 - 30 = 90$.

Following the textbook convention $A_C(z) = \sum_{i=0}^n A_i z^i$, the weight polynomial is

$$A_C(z) = 1 + 30z^2 + 90z^3$$

(equivalently, $W_C(X, Y) = X^3 + 30XY^2 + 90Y^3$ in the homogeneous form).

Problem 2 — MacWilliams formula for $C^\perp$

The MacWilliams identity for a linear $[n, k]$ code over $\mathbb{F}_q$ (cf. codingth.pdf, §14, Theorem 14.2) reads

$$A^\perp(z) = q^{-k} (1 + (q-1)z)^n A\left(\frac{1-z}{1+(q-1)z}\right).$$

With $q = 11$, $n = 3$, $k = 2$:

$$A^\perp(z) = \frac{1}{121} (1 + 10z)^3 A\left(\frac{1-z}{1+10z}\right).$$

Inserting $A(z) = 1 + 30z^2 + 90z^3$, the numerator $(1 + 10z)^3 + 30(1-z)^2(1 + 10z) + 90(1-z)^3$ expands to

$$\begin{aligned} (1 + 10z)^3 &= 1 + 30z + 300z^2 + 1000z^3, \\ 30(1 - z)^2(1 + 10z) &= 30 + 240z - 570z^2 + 300z^3, \\ 90(1 - z)^3 &= 90 - 270z + 270z^2 - 90z^3. \end{aligned}$$

Summing the three lines: $121 + 0 \cdot z + 0 \cdot z^2 + 1210z^3$. Dividing by 121:

$$A^\perp(z) = 1 + 10z^3$$

Interpretation. $C^\perp$ has dimension $n - k = 1$, so $|C^\perp| = 11$. It is generated by the single row (a, b, c) of the check matrix of C . Its codewords are the scalar multiples $\lambda(a, b, c)$ for $\lambda \in \mathbb{F}_{11}$: one zero word (weight 0) and ten nonzero multiples (weight 3, since a, b, c are all nonzero). This is exactly $1 + 10z^3$, confirming the formula.

Problem 3 — $w = (6, 6, 1, 2, 9, 0) \in G$

The code G is the classical Reed–Solomon-type evaluation code

$$G = \{(P(1), P(2), \dots, P(6)) \in \mathbb{F}_{11}^6 \mid P \in \mathbb{F}_{11}[X], \deg P \leq 2\},$$

which has dimension 3. To show $w \in G$ we find $P(X) = \alpha X^2 + \beta X + \gamma \in \mathbb{F}_{11}[X]$ satisfying $w_i = P(i)$ for $i = 1, \dots, 6$.

Use the first three equations:

$$\begin{cases} \alpha + \beta + \gamma \equiv 6, \\ 4\alpha + 2\beta + \gamma \equiv 6, \\ 9\alpha + 3\beta + \gamma \equiv 1. \end{cases} \pmod{11}$$

Subtracting the first from the second: $3\alpha + \beta \equiv 0$.

Subtracting the first from the third: $8\alpha + 2\beta \equiv -5 \equiv 6$, i.e. $4\alpha + \beta \equiv 3$.

Subtracting these two: $\alpha \equiv 3$, hence $\beta \equiv -9 \equiv 2$ and $\gamma \equiv 6 - 3 - 2 = 1$.

So $P(X) = 3X^2 + 2X + 1$. We verify on the remaining points (mod 11):

i	1	2	3	4	5	6
$P(i)$	6	17	34	57	86	121
$P(i) \bmod 11$	6	6	1	2	9	0

All six values match w , so $w = (P(1), \dots, P(6)) \in G$. ■

Problem 4 — Which of R_1, R_2 is unitary?

A real 2×2 matrix M is unitary iff $M^T M = I_2$ iff its columns are orthonormal (over the standard inner product).

Matrix $R_1 = \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}.$

$$R_1^T R_1 = \frac{1}{4} \begin{pmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{pmatrix} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1+3 & -\sqrt{3}+\sqrt{3} \\ -\sqrt{3}+\sqrt{3} & 3+1 \end{pmatrix} = I_2.$$

So R_1 is unitary — it is the rotation $R_{\pi/3}$ (since $\cos(\pi/3) = 1/2$, $\sin(\pi/3) = \sqrt{3}/2$).

Matrix $R_2 = \frac{1}{2} \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix}.$

$$R_2^T R_2 = \frac{1}{4} \begin{pmatrix} \sqrt{2} & \sqrt{2} \\ -\sqrt{2} & \sqrt{2} \end{pmatrix} \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 2+2 & -2+2 \\ -2+2 & 2+2 \end{pmatrix} = I_2.$$

So R_2 is also unitary — it is the rotation $R_{\pi/4}$ (since $\frac{\sqrt{2}}{2} = \cos(\pi/4) = \sin(\pi/4)$).

Conclusion. Both R_1 and R_2 are unitary: each is a real rotation matrix, with $\det R_i = 1$ and orthonormal columns.

Problem 5 — $(A \otimes B)(C \otimes D) = AC \otimes BD$

Let $A \in M_{m \times n}(K)$, $B \in M_{p \times q}(K)$, $C \in M_{n \times r}(K)$, $D \in M_{q \times s}(K)$ so that all the products below are defined.

Proof via simple tensors. The Kronecker product of two linear maps acts on a simple tensor by the rule

$$(A \otimes B)(u \otimes v) = (Au) \otimes (Bv), \quad u \in K^n, v \in K^q.$$

Apply this twice. For any $u \in K^r$, $v \in K^s$,

$$\begin{aligned} (A \otimes B)(C \otimes D)(u \otimes v) &= (A \otimes B)((Cu) \otimes (Dv)) \\ &= A(Cu) \otimes B(Dv) \\ &= (AC)u \otimes (BD)v \\ &= (AC \otimes BD)(u \otimes v). \end{aligned}$$

The simple tensors $\{u \otimes v\}$ span $K^r \otimes K^s$, and both sides are linear; agreement on a spanning set forces equality of the linear maps. Hence $(A \otimes B)(C \otimes D) = AC \otimes BD$. $\square$

Alternative: block computation. Writing $A \otimes B$ as the block matrix with (i, j) -block $a_{ij}B$ (cf. quantumcomp.pdf, §3), the (i, k) -block of the product equals

$$\sum_{j=1}^n (a_{ij}B)(c_{jk}D) = \left(\sum_{j=1}^n a_{ij}c_{jk} \right) BD = (AC)_{ik} BD,$$

which is exactly the (i, k) -block of $AC \otimes BD$.

Problem 6 — Discrete Fourier transform on $\mathbb{Z}_4$

Following the textbook convention (quantumcomp.pdf, §7), the DFT of $f : \mathbb{Z}_n \rightarrow \mathbb{C}$ is

$$\widehat{f}(x) = \frac{1}{\sqrt{n}} \sum_{y \in \mathbb{Z}_n} e^{-\frac{2\pi i xy}{n}} f(y).$$

For $n = 4$ we have $e^{-2\pi i/4} = e^{-i\pi/2} = -i$, so

$$\widehat{f}(x) = \frac{1}{2} \sum_{y=0}^3 (-i)^{xy} f(y).$$

Writing out the four terms $y = 0, 1, 2, 3$, and using $(-i)^{2x} = (-1)^x$ and $(-i)^{3x} = i^x$:

$$\widehat{f}(x) = \frac{1}{2} \left[f(0) + (-i)^x f(1) + (-1)^x f(2) + i^x f(3) \right]$$

Evaluating at each $x \in \mathbb{Z}_4$:

$$\begin{aligned} \widehat{f}(0) &= \frac{1}{2} [f(0) + f(1) + f(2) + f(3)], \\ \widehat{f}(1) &= \frac{1}{2} [f(0) - i f(1) - f(2) + i f(3)], \\ \widehat{f}(2) &= \frac{1}{2} [f(0) - f(1) + f(2) - f(3)], \\ \widehat{f}(3) &= \frac{1}{2} [f(0) + i f(1) - f(2) - i f(3)]. \end{aligned}$$

These four rows are exactly the rows of the unitary 4×4 DFT matrix

$$F_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$